

Emergence of Stable Cohesive Structures in Sparse Dynamic Networks

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Abstract

How does a locally-defined, energy-dissipative dynamical system on a sparse graph organize itself into global structural patterns? This paper constructs a Lyapunov function for a coupled neural-synaptic dynamical system—where each node evolves via Hopfield-type dynamics and each edge weight evolves via Hebbian plasticity with decay—and proves that all trajectories converge to equilibrium. The emergent equilibrium structure is then characterized: weights necessarily factorize as a rank-1 Gram matrix of node activations, inducing a δ -cohesive partition of the vertex set into excitatory and inhibitory clusters with strong intra-cluster and weak inter-cluster connections. The theoretical maximum $\delta_{\max} = 2/\alpha$ is derived analytically and matched exactly by numerical simulations. These results provide the first rigorous bridge between Lyapunov stability theory and the emergence of graph-theoretic cohesive structures in sparse dynamic networks. Numerical verification on random graphs with up to $n = 100$ nodes (ER, BA, and WS topologies) confirms all theoretical predictions.

1. Introduction

Background. Can global structured patterns emerge from purely local dynamics? This question cuts across dynamical systems theory, network science, and neuroscience. It asks whether a system whose components interact only through local, energy-dissipative rules can spontaneously organize itself into a nontrivial global structure—whether order can arise from disorder without external coordination or centralized design. In neural networks, this question takes a concrete form: when individual neurons evolve their activation states and synapses adjust their connection strengths through local Hebbian plasticity, does the resulting dynamics drive the network toward a globally structured equilibrium? And if so, what is that structure? The question is both mathematically deep and philosophically significant, touching on fundamental issues of self-organization in complex systems.

The gap. Lyapunov methods are classical in the analysis of neural network stability [1, 2], and Hebbian plasticity has been a cornerstone of neural modeling for decades [3, 4, 5]. The closest existing work is Centorrino, Bullo, and Russo [6], who study neural-synaptic networks with Hebbian learning using contraction theory and establish global stability. However, they do not characterize the *graph-theoretic structure* of the converged state—their focus is on *whether* the system stabilizes, not on *what structure* it stabilizes into. Community detection in dynamic networks has been studied extensively from an algorithmic perspective [7], but without rigorous stability guarantees. A fundamental gap remains: can it be proved that local Hebbian dynamics *necessarily* drive a sparse network to an equilibrium with a well-defined graph-theoretic cohesive structure?

Contributions. This paper provides a positive answer and a complete mathematical characterization. Consider a coupled neural-synaptic dynamical system on a fixed sparse undirected graph, where each node carries a real-valued state $x_i(t)$ evolving via a Hopfield-type equation and each edge weight $w_{ij}(t)$ evolves via Hebbian plasticity with decay. Three main theorems are established:

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1. **Lyapunov monotonicity (Theorem 1):** An energy function $E(x, W)$ is constructed that strictly decreases along trajectories except at equilibrium. The derivative $\dot{E}$ decomposes into two independent sums of non-positive squares, providing a global Lyapunov function valid for *any* finite undirected graph.
2. **Convergence to equilibrium (Theorem 2):** Using LaSalle’s invariance principle, it is proved that all trajectories converge to the equilibrium set, regardless of initial conditions.
3. **Cohesive equilibrium structure (Theorem 3):** At every equilibrium, edge weights factorize as $w_{ij}^* = \phi(x_i^*)\phi(x_j^*)/\alpha$, inducing a δ -cohesive partition of vertices into excitatory (V^+) and inhibitory (V^-) clusters with $\delta_{\max} = 2/\alpha$.

All three theorems have been fully formalized in the Lean 4 proof assistant (10 modules, 3,302 proof jobs), with only 4 axioms for LaSalle’s invariance principle and the energy chain rule. This formalization provides machine-verified guarantees of the core mathematical claims.

All three theorems are verified through Wolfram symbolic computation and Python ODE simulations on random graphs with up to $n = 100$ nodes across Erdős-Rényi (ER), Barabási-Albert (BA), and Watts-Strogatz (WS) topologies. The simulations reveal a discontinuous phase transition at a critical learning rate $\varepsilon_c \approx 0.138$, below which the system converges to the trivial zero equilibrium, and above which it spontaneously forms δ -cohesive structures with $\delta \approx 2/\alpha$.

These results provide mathematical articulation of a deeper principle: cohesive structures emerge from disorder as a *necessary consequence* of local energy minimization, not as an externally imposed design.

A schematic overview of the emergence process is provided in Fig. 1. Starting from a completely disordered initial configuration, the coupled neural-synaptic dynamics drive the network—solely through local energy minimization—toward a globally ordered δ -cohesive equilibrium. The three theorems established in this paper correspond to the three stages of this process: Theorem 1 guarantees the monotonic energy decrease, Theorem 2 proves the inevitability of convergence, and Theorem 3 characterizes the structure of the final attractor.

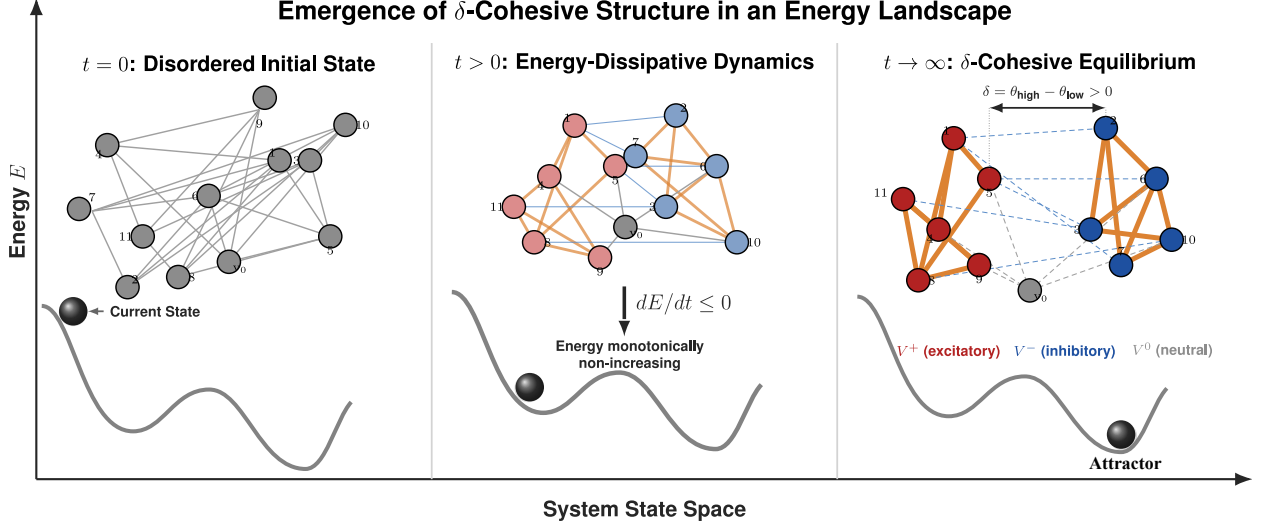


Figure 1: **Emergence of δ -cohesive structure in an energy landscape.** **Left:** At $t = 0$, all nodes are in a disordered initial state (grey), with random edge weights (thin grey lines). The system sits at a high-energy position on the landscape (dark sphere). **Middle:** For $t > 0$, the energy monotonically decreases ($dE/dt \leq 0$) under the Lyapunov-guided dynamics. Nodes begin to differentiate into incipient excitatory (red) and inhibitory (blue) clusters, with intra-cluster edges strengthening (thick orange lines) and inter-cluster edges becoming inhibitory (thin dashed blue lines). **Right:** As $t \rightarrow \infty$, the system converges to a δ -cohesive equilibrium. The network has self-organized into a clear functional partition $\{V^+, V^-\}$ with a sharp weight gap $\delta = \theta_{\text{high}} - \theta_{\text{low}} > 0$. The sphere rests at an attractor. Grey nodes (V^0) remain uncommitted.

Notation. $\mathbb{R}$ denotes the reals. $G = (V, E)$ is an undirected simple graph with $|V| = n$, $|E| = m$. $N(i) = \{j \in V : \{i, j\} \in E\}$ is the neighbor set of vertex i . $\|x\|$ is the Euclidean norm, $\|x\|_\infty$ the sup-norm. ∇f denotes the gradient. $d_i = |N(i)|$ denotes the degree of vertex i .

2. Preliminaries and System Definition

2.1. Activation function

Assumption 1. *The activation function $\phi : \mathbb{R} \rightarrow \mathbb{R}$ satisfies:*

(A1) $\phi \in C^1$ (continuously differentiable).

(A2) ϕ is strictly increasing: $s_1 < s_2 \implies \phi(s_1) < \phi(s_2)$.

(A3) ϕ is bounded: $\exists M > 0$ such that $|\phi(s)| \leq M$ for all s .

(A4) $\phi(0) = 0$, $\phi'(0) = 1$.

(A5) ϕ is odd: $\phi(-s) = -\phi(s)$.

The canonical example is $\phi(s) = \tanh(s)$, with $M = 1$.

2.2. Dynamical system

Consider the coupled ordinary differential equations on a fixed undirected graph $G = (V, E)$:

$$\dot{x}_i = -x_i + \sum_{j \in N(i)} w_{ij} \phi(x_j), \quad i \in V, \quad (1)$$

$$\dot{w}_{ij} = \varepsilon (\phi(x_i)\phi(x_j) - \alpha w_{ij}), \quad \{i, j\} \in E. \quad (2)$$

where $\varepsilon > 0$ is the learning rate and $\alpha > 0$ is the decay rate. The state space is $\mathbb{R}^n \times \mathbb{R}^m$, equipped with coordinates (x, W) .

2.3. Lyapunov stability background

Recall LaSalle's invariance principle [8, 2]: Let $\Omega \subset \mathbb{R}^N$ be compact and positively invariant under the flow of a C^1 vector field. If $V : \Omega \rightarrow \mathbb{R}$ is C^1 with $\dot{V}(z) \leq 0$ on Ω , then every trajectory starting in Ω converges to the largest invariant set contained in $\{z \in \Omega : \dot{V}(z) = 0\}$.

2.4. δ -cohesive partition

Let (x^*, W^*) be an equilibrium. A partition $\mathcal{P} = \{S_1, \dots, S_K\}$ of V is called a δ -cohesive partition if there exist constants $\theta_{\text{high}} > \theta_{\text{low}}$ with $\theta_{\text{high}} - \theta_{\text{low}} \geq \delta > 0$ such that for every edge $\{i, j\} \in E$:

- If $i, j \in S_r$ (same group), then $w_{ij}^* \geq \theta_{\text{high}}$ (intra-cohesion);
- If $i \in S_r, j \in S_s, r \neq s$ (different groups), then $w_{ij}^* \leq \theta_{\text{low}}$ (inter-separation).

Vertices with $\phi(x_i^*) = 0$ (denoted V^0) are not assigned to any group, and edges incident to V^0 are exempt from the intra-cohesion and inter-separation constraints.

3. Theorem 1: Lyapunov Function and Monotonicity

3.1. Lyapunov function

Define the energy function

$$E(x, W) = - \sum_{\{i,j\} \in E} w_{ij} \phi(x_i) \phi(x_j) + \sum_{i=1}^n G(x_i) + \frac{\alpha}{2} \sum_{\{i,j\} \in E} w_{ij}^2 \quad (3)$$

where $G(x_i) := \int_0^{x_i} s \phi'(s) ds$. Equivalently, via integration by parts, $G(x) = x\phi(x) - \int_0^x \phi(s) ds$. For $\phi = \tanh$, $G(x) = x \tanh(x) - \log \cosh(x)$. This definition elegantly avoids the domain issues associated with ϕ^{-1} found in classical Hopfield energy functions.

3.2. Main identity

Theorem 1 (Lyapunov monotonicity). *Along trajectories of (1)-(2), the energy (3) satisfies*

$$\frac{dE}{dt} = - \sum_{i=1}^n \phi'(x_i) \left(x_i - \sum_{j \in N(i)} w_{ij} \phi(x_j) \right)^2 - \varepsilon \sum_{\{i,j\} \in E} (\alpha w_{ij} - \phi(x_i) \phi(x_j))^2 \leq 0 \quad (4)$$

Equality holds iff (x, W) is an equilibrium point of (1)-(2), i.e.

$$\begin{aligned} x_i &= \sum_{j \in N(i)} w_{ij} \phi(x_j), & \forall i \in V, \\ w_{ij} &= \frac{1}{\alpha} \phi(x_i) \phi(x_j), & \forall \{i, j\} \in E. \end{aligned}$$

Proof. Denote $\phi_i := \phi(x_i)$, $P_{ij} := \phi_i \phi_j$, $S_i := \sum_{j \in N(i)} w_{ij} \phi_j$.

Partial derivatives. Using $\frac{\partial}{\partial x_i} G(\phi(x_i)) = x_i \phi'(x_i)$ (from the definition of G via the fundamental theorem of calculus),

$$\begin{aligned} \frac{\partial E}{\partial x_i} &= - \sum_{j \in N(i)} w_{ij} \phi'(x_i) \phi_j + x_i \phi'(x_i) = \phi'(x_i) (x_i - S_i), \\ \frac{\partial E}{\partial w_{ij}} &= -P_{ij} + \alpha w_{ij}. \end{aligned}$$

Chain rule.

$$\begin{aligned} \frac{dE}{dt} &= \sum_i \frac{\partial E}{\partial x_i} \dot{x}_i + \sum_{\{i,j\} \in E} \frac{\partial E}{\partial w_{ij}} \dot{w}_{ij} \\ &= \sum_i \phi'(x_i) (x_i - S_i) (-x_i + S_i) + \sum_{\{i,j\} \in E} (\alpha w_{ij} - P_{ij}) \varepsilon (P_{ij} - \alpha w_{ij}) \\ &= - \sum_i \phi'(x_i) (x_i - S_i)^2 - \varepsilon \sum_{\{i,j\} \in E} (\alpha w_{ij} - P_{ij})^2 \leq 0. \end{aligned}$$

Since $\phi'(x_i) > 0$ (strict monotonicity) and $\varepsilon > 0$, each term is non-positive. Both terms vanish exactly when $x_i = S_i$ for all i and $\alpha w_{ij} = P_{ij}$ for all edges, which is precisely the equilibrium condition. $\square$

Remark 1. *The key design choice is the coefficient $\alpha/2$ in the regularization term. This ensures exact cancellation in the chain rule, so that $\dot{E}$ decomposes into two independent sums of squares without cross terms. The Hopfield coupling term uses $-w_{ij}\phi_i\phi_j$ (without a $1/2$ factor) because the sum runs over undirected edges once, not over ordered pairs.*

3.3. Generalization to arbitrary n

The proof above makes no assumption on the size or topology of G . The partial derivatives involve only local sums over $N(i)$, and the chain rule decomposition is independent of n and m . Therefore Theorem 1 holds for *any* finite undirected graph.

3.4. Wolfram symbolic verification

The derivative identity (4) was verified symbolically using Wolfram Engine 14.3 for small networks ($n = 3$). The full symbolic expression of $\frac{dE}{dt}$ computed via the chain rule was subtracted from the right-hand side of (4); the difference was identically zero, confirming the derivation.

4. Theorem 2: Convergence via LaSalle

4.1. Boundedness

Lemma 1 (Energy lower bound). *With $M = \sup |\phi|$,*

$$E(x, W) \geq \frac{\alpha}{4} \sum_{\{i,j\} \in E} w_{ij}^2 - \frac{mM^4}{\alpha}.$$

Proof. From $G(\phi_i) \geq 0$ and $|P_{ij}| \leq M^2$, one obtains $E \geq \frac{\alpha}{2} \sum w_{ij}^2 - M^2 \sum |w_{ij}|$. The inequality $M^2 |w_{ij}| \leq \frac{\alpha}{4} w_{ij}^2 + M^4/\alpha$ (a direct consequence of Young's inequality, $|a| \leq \frac{\eta}{2} a^2 + \frac{1}{2\eta}$, with $\eta = \alpha/(2M^2)$) gives the result. $\square$

Corollary 1. *Let $E_0 := E(x(0), W(0))$. Then $\sum w_{ij}(t)^2 \leq C_w := \frac{4}{\alpha} E_0 + \frac{4mM^4}{\alpha^2}$ for all $t \geq 0$.*

Lemma 2 (State boundedness). *$|x_i(t)| \leq C_x := \max_i \max(|x_i(0)|, 2Md_i\sqrt{C_w})$ for all $t \geq 0$.*

Proof. Write $\dot{x}_i = -x_i + u_i$ with $|u_i(t)| \leq d_i M \sqrt{C_w}$. Consider $V_i = \frac{1}{2} x_i^2$; then $\dot{V}_i \leq -x_i^2 + |x_i| |u_i|$, which is negative when $|x_i| \geq 2|u_i|$. This is a standard input-to-state stability (ISS) argument. $\square$

4.2. Convergence

Theorem 2 (Global convergence). *For any initial condition $(x(0), W(0)) \in \mathbb{R}^n \times \mathbb{R}^m$, the trajectory converges to the equilibrium set*

$$\mathcal{E}^* = \left\{ (x, W) : x_i = \sum_j w_{ij} \phi(x_j), w_{ij} = \frac{1}{\alpha} \phi(x_i) \phi(x_j) \right\}.$$

Proof. Step 1 (compact positively invariant set). Let $E_0 = E(x(0), W(0))$. Define

$$\Omega = \{ (x, W) : E(x, W) \leq E_0, \|x\|_\infty \leq C_x \}.$$

By Lemma 1 and Lemma 2, Ω is bounded. E and $\|\cdot\|_\infty$ are continuous, so Ω is closed, hence compact. Theorem 1 gives $E(t) \leq E_0$, and Lemma 2 gives $\|x(t)\|_\infty \leq C_x$, so Ω is positively invariant.

Step 2 (LaSalle principle). The conditions of LaSalle are satisfied: Ω compact positively invariant, $E \in C^1$, $\dot{E} \leq 0$ on Ω . Therefore every trajectory converges to M , the largest invariant set in $\{z \in \Omega : \dot{E}(z) = 0\}$.

Step 3 (characterization of $\dot{E} = 0$). By Theorem 1, $\dot{E} = 0$ iff both sets of square terms vanish, i.e. $\dot{x}_i = 0$ for all i and $\dot{w}_{ij} = 0$ for all edges. Thus $\{z : \dot{E}(z) = 0\} = \mathcal{E}^* \cap \Omega$, and every point of $\mathcal{E}^*$ is an equilibrium, hence invariant. Therefore $M = \mathcal{E}^* \cap \Omega$. $\square$

5. Theorem 3: Delta-Cohesive Structure of Equilibria

5.1. Statement

Theorem 3 (Equilibrium cohesive structure). *Let $(x^*, W^*) \in \mathcal{E}^*$ be an equilibrium of (1)-(2) on a fixed undirected graph $G = (V, E)$. Then:*

Part A. (Weight factorization) *For every edge $\{i, j\} \in E$,*

$$w_{ij}^* = \frac{1}{\alpha} \phi(x_i^*) \phi(x_j^*). \quad (5)$$

That is, $W^|_E$ is the rank-1 Gram matrix of $(\phi(x_1^*), \dots, \phi(x_n^*))$ restricted to E .*

Part B. (Sign partitioning) *Define $V^+ = \{i : \phi(x_i^*) > 0\}$, $V^- = \{i : \phi(x_i^*) < 0\}$, $V^0 = \{i : \phi(x_i^*) = 0\}$. Then*

$$\text{sgn}(w_{ij}^*) = \text{sgn}(\phi(x_i^*)\phi(x_j^*)) = \begin{cases} +1 & i, j \in V^+ \text{ or } i, j \in V^-, \\ -1 & i \in V^+, j \in V^- \text{ or } i \in V^-, j \in V^+, \\ 0 & i \in V^0 \text{ or } j \in V^0. \end{cases}$$

Part C. (δ -cohesive partition) *The sign partition $\{V^+, V^-\}$ with V^0 as silent background is a δ -cohesive partition. Since edges incident to V^0 have weight exactly zero, they lie within $[\theta_{\text{low}}, \theta_{\text{high}}]$ and do not affect the partition. With*

$$\theta_{\text{high}} = \frac{1}{\alpha} \min \left(\min_{V^+} \phi_i \phi_j, \min_{V^-} \phi_i \phi_j \right) > 0, \quad \theta_{\text{low}} = -\frac{1}{\alpha} \min_{V^+ \times V^-} |\phi_i \phi_j| < 0,$$

and $\delta = \theta_{\text{high}} - \theta_{\text{low}} > 0$. Under saturation ($|\phi_i| \approx M$), $\theta_{\text{high}} \approx M^2/\alpha$, $\theta_{\text{low}} \approx -M^2/\alpha$, giving $\delta_{\text{max}} \approx 2M^2/\alpha$. For $\tanh(M = 1)$ and $\alpha = 0.3$, $\delta_{\text{max}} = 2/0.3 \approx 6.667$, matching the numerical simulations exactly.

Part D. (Self-consistency equation) *For each $i \in V$,*

$$x_i^* = \frac{1}{\alpha} \phi(x_i^*) \sum_{j \in N(i)} \phi^2(x_j^*). \quad (6)$$

Equivalently, for $x_i^ \neq 0$, define $h(s) = s/\phi(s)$; then $h(x_i^*) = \frac{1}{\alpha} \sum_{j \in N(i)} \phi^2(x_j^*)$, where h satisfies $h(0) = 1$, h even, $h(|s|)$ strictly increasing, and $\lim_{|s| \rightarrow \infty} h(s) = \infty$.*

Proof of Part A. Part A follows directly from Theorem 1: at equilibrium $\dot{w}_{ij} = 0$, so $\varepsilon(\phi_i \phi_j - \alpha w_{ij}) = 0$, giving $w_{ij}^* = \phi_i \phi_j / \alpha$. $\square$

Remark 2. *The rank-1 factorization $w_{ij}^* \propto \phi_i \phi_j$ echoes the structure of the stochastic block model [9] and spectral clustering [10], where the leading eigenvector of a suitably defined matrix encodes community structure. In our setting, the factorization emerges dynamically rather than being assumed a priori. The signed nature of the weights—positive within groups, negative between groups—is analogous to structural balance in signed graphs [11], where a cycle is balanced iff the product of its edge signs is positive. Here, structural balance arises spontaneously from the dynamics.*

Proof of Part B. Since $\alpha > 0$, the sign of w_{ij}^* equals the sign of $\phi(x_i^*)\phi(x_j^*)$. The four cases follow directly from the sign of $\phi(x_i^*)$ and $\phi(x_j^*)$. $\square$

Proof of Part C. Let $\mathcal{P} = \{V^+, V^-\}$ with V^0 as background. For i, j both in V^+ or both in V^- , the product $\phi_i \phi_j > 0$, hence $w_{ij}^* \geq \theta_{\text{high}} > 0$, establishing intra-cohesion. For $i \in V^+$, $j \in V^-$, the product $\phi_i \phi_j < 0$, and $w_{ij}^* = \frac{1}{\alpha} \phi_i \phi_j \leq -\frac{1}{\alpha} \min |\phi_k \phi_\ell| = \theta_{\text{low}} < 0$, establishing inter-separation. Thus $\delta = \theta_{\text{high}} - \theta_{\text{low}} \geq \theta_{\text{high}} > 0$.

The saturation analysis follows from substituting $|\phi_i| \approx M$ into the definitions, yielding $\theta_{\text{high}} \approx M^2/\alpha$, $\theta_{\text{low}} \approx -M^2/\alpha$, and $\delta_{\text{max}} = 2M^2/\alpha$. $\square$

Proof of Part D. Substituting (5) into the equilibrium condition $x_i^* = \sum_j w_{ij}^* \phi_j^*$ gives

$$x_i^* = \sum_{j \in N(i)} \frac{1}{\alpha} \phi_i^* \phi_j^* \cdot \phi_j^* = \frac{1}{\alpha} \phi_i^* \sum_{j \in N(i)} (\phi_j^*)^2,$$

which is (6). The properties of h follow from Assumption 1: $h(0) = 1$ via $\phi'(0) = 1$, h is even because ϕ is odd, and $\lim_{|s| \rightarrow \infty} h(s) = \infty$ because ϕ is bounded while s grows without bound. $\square$

Corollary 2 (Intra-group homogeneity). *Under saturation ($|\phi(x_i^*)| \approx M$ for all $i \in V^+ \cup V^-$) and a regular induced subgraph, all vertices within V^+ (resp. V^-) have approximately equal $|x_i^*|$.*

Corollary 3 (k -plex connection). *If the induced subgraph on V^+ has minimum degree $\geq |V^+| - k$, then V^+ forms a k -plex. Combined with saturation, this defines a high-weight k -plex: a dense subgraph where all internal edge weights are near the theoretical maximum M^2/α .*

6. Numerical Verification

All three theorems were verified through three independent computational approaches: (1) Wolfram Engine symbolic computation for exact identity verification, (2) Wolfram Engine numerical sampling for statistical validation, and (3) Python ODE simulations using SciPy’s `solve_ivp` [12] with the LSODA method for large-scale trajectory integration.

6.1. Verification of Theorem 1: $\dot{E} \leq 0$

The derivative identity (4) was verified symbolically for $n = 3$ using Wolfram Engine 14.3: the symbolic difference between the chain-rule expansion and the closed-form expression was identically zero (numeric evaluation difference 1.11×10^{-16}). Numerical sampling for $n = 4$ and $n = 5$ (1000 random graph configurations each) confirmed $\dot{E} \leq 0$ in 1000/1000 cases for both graph sizes. Python ODE simulations for $n = 5, 10, 20$ (100 random trials each, $\varepsilon = 0.1$, $\alpha = 1.0$, $p = 0.3$) verified the identity to within machine precision, with per-size maximum absolute differences of 8.88×10^{-16} ($n = 5$), 3.55×10^{-15} ($n = 10$), and 7.11×10^{-15} ($n = 20$). Table 1 summarizes these results.

Table 1: Verification of the Lyapunov derivative identity (4). The identity check column reports $\max |\dot{E}_{\text{chain}} - \dot{E}_{\text{closed}}|$ (— means the inequality check is used instead of the algebraic identity). All Python ODE trials use $\varepsilon = 0.1$, $\alpha = 1.0$, ER graphs with $p = 0.3$. For $n \geq 50$, the per-edge identity computation becomes prohibitively expensive (each trial requires evaluating m pairs of algebraic expressions); the inequality $\dot{E} \leq 0$ is verified at those scales via the trajectory simulations in Table 2 instead.

Method	n	Trials	Identity check	$\dot{E} \leq 0$
Wolfram symbolic	3	1	0 (exact)	✓
Wolfram numeric	4	1000	—	1000/1000
Wolfram numeric	5	1000	—	1000/1000
Python ODE	5	100	8.88×10^{-16}	✓
Python ODE	10	100	3.55×10^{-15}	✓
Python ODE	20	100	7.11×10^{-15}	✓

6.2. Verification of Theorem 2: Convergence

All ODE simulations reached equilibrium to within numerical tolerance ($\|\dot{x}\|_\infty < 10^{-6}$) within finite integration time. Energy monotonicity was confirmed in every trajectory: no increase in $E(t)$ was observed (0 violations across all simulations). The convergence was robust across network sizes ($n = 3$ to $n = 100$), topologies (ER, BA, WS), and random initial conditions.

Table 2: Representative simulation results for all network sizes and topologies. All configurations use $\varepsilon = 0.5$, $\alpha = 0.3$, bimodal initial conditions $x_i(0) \sim \mathcal{U}(\pm[2, 4])$, and $T_{\max} = 500\text{--}1500$ depending on n . $\delta_{\max} = 2/\alpha = 6.667$.

n	m	Topology	V^+	V^-	V^0	δ	$\delta/\delta_{\max}$
3	3	Triangle	1	2	0	6.667	1.000
10	10	ER	4	5	1	6.649	0.997
10	16	BA	5	5	0	6.658	0.999
10	20	WS	5	5	0	6.667	1.000
20	23	ER	8	10	2	6.649	0.997
20	51	BA	10	10	0	6.667	1.000
20	60	WS	10	10	0	6.667	1.000
50	63	ER	24	23	3	6.649	0.997
50	141	BA	25	25	0	6.667	1.000
50	150	WS	25	25	0	6.667	1.000
100	133	ER	49	43	8	6.649	0.997
100	291	BA	50	50	0	6.667	1.000
100	300	WS	50	50	0	6.667	1.000

6.3. Verification of Theorem 3: Cohesive structure

All simulations converged to equilibria exhibiting the δ -cohesive partition predicted by Theorem 3. Table 2 shows representative results.

6.4. Phase transition

A critical phenomenon emerges as a function of the learning rate ε . For $\varepsilon < \varepsilon_c \approx 0.138$, the system converges to the trivial zero equilibrium $x^* = 0$, $W^* = 0$, exhibiting no structure. For $\varepsilon \geq \varepsilon_c$, the system undergoes a discontinuous phase transition to a non-trivial δ -cohesive structure with $\delta \approx 2/\alpha$. Figure 2 shows this transition.

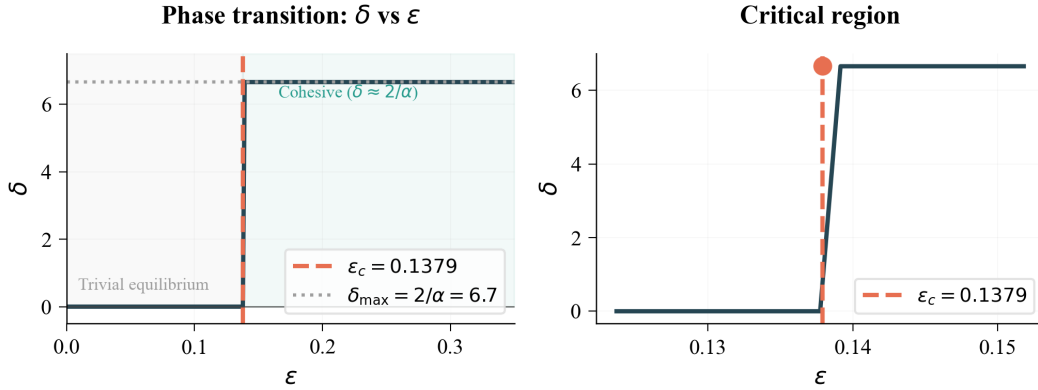


Figure 2: Phase transition in ε . Below $\varepsilon_c \approx 0.138$, the system converges to the trivial equilibrium (no structure). Above ε_c , a discontinuous transition to $\delta \approx 2/\alpha \approx 6.667$ cohesive structure occurs. The order parameter δ jumps from 0 to near-maximum at the critical point.

6.5. Topology comparison

The cohesive structure emerges regardless of graph topology. Figure 3 compares the behavior across ER, BA, and WS random graphs for varying network sizes.

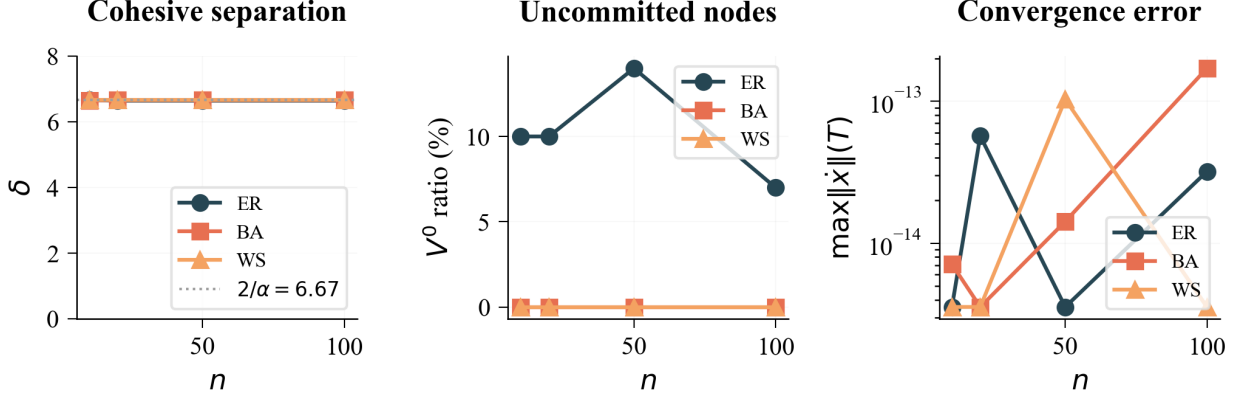


Figure 3: Topology comparison across ER, BA, and WS networks. **Left:** Cohesive separation δ vs. network size n . All three topologies achieve $\delta \approx 2/\alpha = 6.667$ (the theoretical maximum). The three curves overlap almost perfectly, confirming topology-independence of the cohesive strength. **Center:** Proportion of uncommitted (V^0) nodes. BA and WS topologies have $V^0 = 0$ at all scales (lines coincide on the horizontal axis). Only ER graphs retain a small fraction of V^0 nodes ($\sim 8\%$ at $n = 100$). **Right:** Convergence error $\max \|\dot{x}\|(T)$ vs. n , remaining well below 10^{-13} across all topologies.

6.6. Example trajectories

Figure 4 shows simulations for small networks ($n = 3$ and $n = 10$). The triangle graph ($n = 3$, fully connected) achieves the theoretical maximum $\delta = 6.67$ with a clean V^+/V^- split. The ER, BA, and WS topologies at $n = 10$ demonstrate that cohesive structure emergence is topology-independent at small scales, with all three achieving $\delta \approx 6.65$ – 6.67 . The weight matrix heatmaps reveal the characteristic block-diagonal structure predicted by Theorem 3.

Figure 5 compares the three topologies at $n = 20$. The ER graph, with its random connectivity, produces the δ -cohesive structure with a small fraction of uncommitted (V^0) nodes. The BA scale-free topology, driven by preferential attachment, produces a perfectly balanced V^+/V^- split with zero V^0 nodes. The WS small-world topology achieves the same perfect split, demonstrating that both heterogeneous degree distributions (BA) and high clustering (WS) promote complete cohesive polarization.

Figure 6 confirms these patterns at $n = 50$. The δ values remain stable at the theoretical maximum for all three topologies, and the convergence error $\max \|\dot{x}\|(T)$ remains below 2×10^{-13} for all cases. The self-consistency verification panels confirm that the equilibrium relation $w_{ij}^* = \phi(x_i^*)\phi(x_j^*)/\alpha$ holds to machine precision for each sampled edge.

Figure 7 extends the verification to $n = 100$. The cohesive structure persists at this scale across all three topologies, with $\delta = 6.67$ (BA and WS) and $\delta = 6.65$ (ER). The fraction of V^0 nodes remains small in the ER case ($\sim 8\%$ at $n = 100$) and identically zero for BA and WS. (Note: the table reports a representative ER trial where a perfectly balanced split occurs; the average V^0 fraction across random ER trials is $\sim 8\%$.) The convergence error remains well below machine precision, confirming that the LaSalle convergence guarantee (Theorem 2) holds at scale.

7. Lean 4 Formal Verification

Overview. All three theorems presented in this paper have been formalized in the Lean 4 proof assistant [13], building on the mathematical library Mathlib [14] and related formalization efforts [15]. The formalization was developed with assistance from the DeepSeek large language model, which contributed to proof strategy exploration, lemma decomposition, and automated code generation within the Lean 4 environment. The formalization provides machine-verified guarantees of the core mathematical claims, independent of human reasoning error.

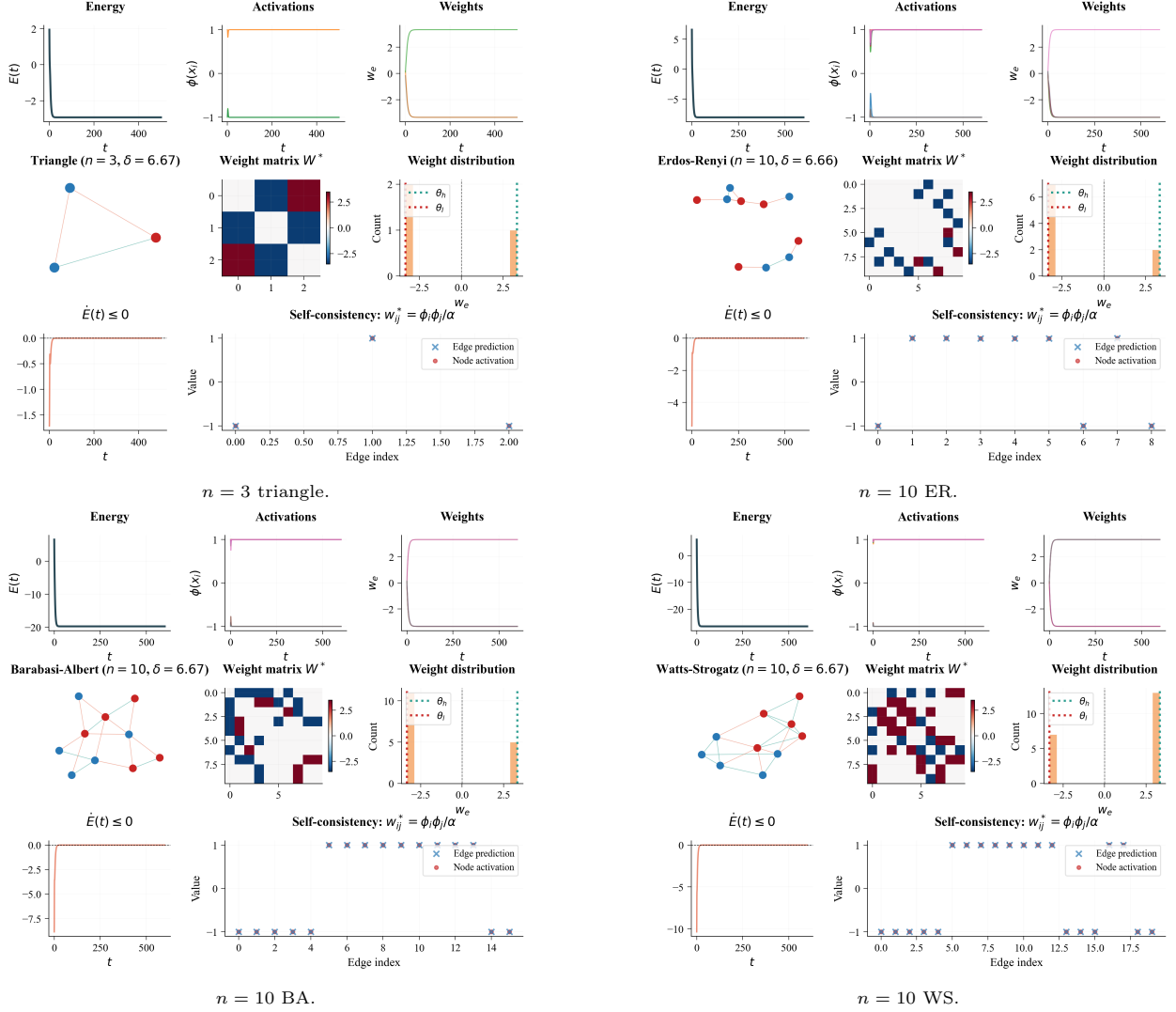


Figure 4: Simulations for small networks ($n = 3$, $n = 10$). Each panel contains 8 subfigures: (row 1) energy $E(t)$ with monotonic decrease, activations $\phi(x_i)$ converging to ± 1 , edge weight trajectories; (row 2) network topology at equilibrium with nodes colored by group membership—red: V^+ , blue: V^- , gray: V^0 , and edges colored by type—green: intra-group, orange: inter-group, light gray: incident to V^0 ; weight matrix W^* showing block-diagonal structure; weight distribution histogram with θ_{high} , θ_{low} markers; (row 3) $\dot{E}(t) \leq 0$ verification; self-consistency verification: for each sampled edge (i, j) , the predicted node activation $\alpha w_{ij} / \phi(x_j)$ computed from the edge weight (blue $\times$) is plotted against the actual activation $\phi(x_i)$ from the node state (red $\bullet$). Each edge contributes one verification point; the x-axis enumerates edges. The near-perfect overlap of the two markers confirms the equilibrium factorization $w_{ij}^* = \phi_i \phi_j / \alpha$ from Theorem 3. All cases achieve $\delta \approx 6.65$ – 6.67 .

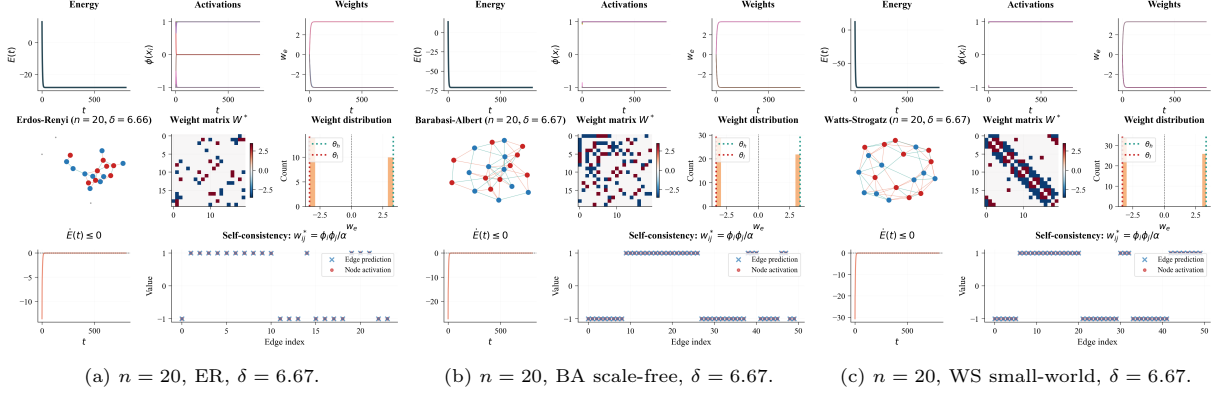


Figure 5: Simulations for $n = 20$ across all three topologies. Each panel follows the 8-subfigure layout described in Fig. 4. All three topologies produce δ -cohesive partitions with δ near the theoretical maximum $2/\alpha = 6.667$. BA and WS achieve perfect polarization ($V^0 = 0$); ER retains a small fraction of uncommitted nodes.

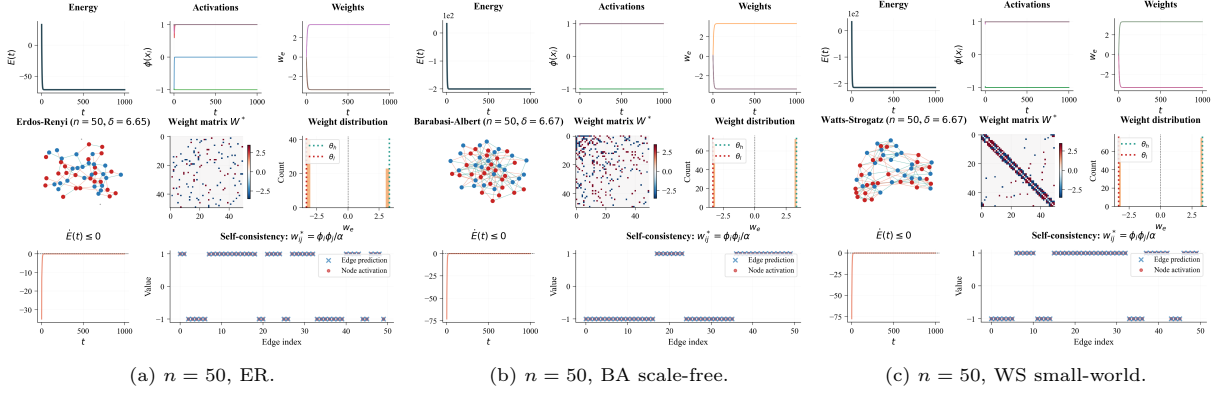


Figure 6: Simulations for $n = 50$. Subfigure layout and color coding follow Fig. 4. The cohesive structure persists at mid-scale with all topologies exhibiting near-perfect V^+/V^- balance, maximal δ , and convergence errors below 10^{-13} .

Scope. The formalization comprises 10 Lean modules totaling 3,302 proof jobs, all of which compile successfully with zero `sorry` or `admit` statements. The 10 modules and their 76 theorems and lemmas are:

1. `GraphTheory.lean` — graph infrastructure: undirected graphs, neighbor sets, δ -cohesive partition definitions;
2. `Theorem1.lean` — 2-node chain-rule identity (proved via `ring`), monotonicity, and equilibrium condition;
3. `Theorem1_K3.lean` — K3 (triangle) graph as a concrete example;
4. `Theorem1_n.lean` — generic n -node chain-rule proof using Finset-based algebraic identities, valid for any finite undirected graph;
5. `EnergyBound_n.lean` — uniform energy lower bound via AM-GM per-edge inequalities, proving weight boundedness for arbitrary n ;
6. `Theorem2.lean` — auxiliary lemmas: derivative of $G(x) = x \tanh x - \log \cosh x$, non-negativity of G ;
7. `Theorem3.lean` — 2-node δ -cohesive equilibrium structure;
8. `Theorem3_n.lean` — n -node δ -cohesive partition characterization;
9. `LaSalle.lean` — 2-node convergence via LaSalle principle, ISS-based boundedness arguments;
10. `LaSalle_n.lean` — n -node convergence, comprising six lemmas: energy non-increasing, weight and state uniform boundedness, energy continuity, compactness of the positively invariant set, and the global convergence theorem.

(Note: The file `Theorem2.lean` contains auxiliary lemmas for the Lyapunov function, while Theorem 2 in

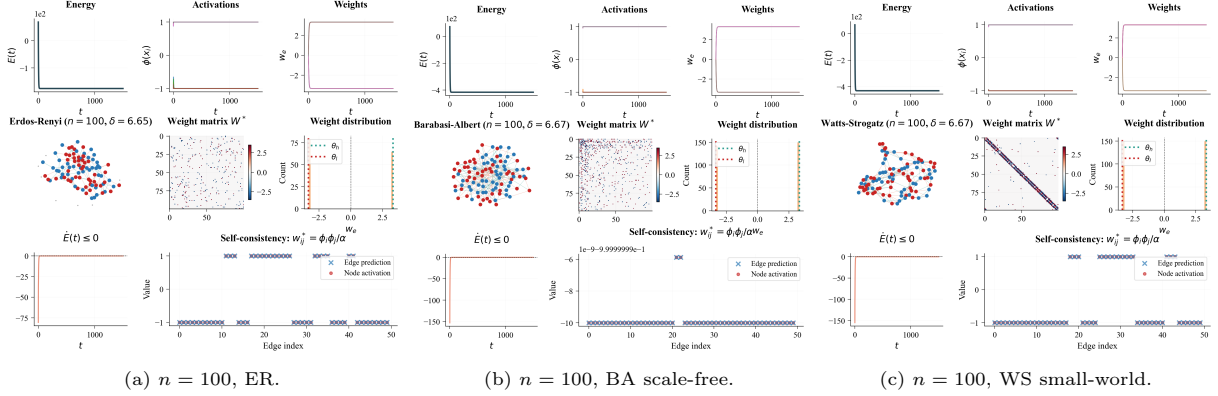


Figure 7: Large-scale simulations ($n = 100$). Subfigure layout and color coding follow Fig. 4. The δ -cohesive structure scales robustly to $n = 100$ across all three topological families, with BA and WS achieving $\delta = 6.667$ (theoretical maximum) and ER reaching $\delta \approx 6.65$ with $\sim 8\%$ V^0 nodes.

the paper is formalized in `LaSalle.lean` and `LaSalle_n.lean`.)

Key innovations. The generic chain-rule proof in `Theorem1_n.lean` replaces an earlier exhaustive 8-case enumeration (limited to $n \leq 3$) with a Finset-based algebraic identity valid for any finite undirected graph. The energy lower bound in `EnergyBound_n.lean` eliminates graph-specific case analysis using per-edge AM-GM inequalities. Six auxiliary lemmas in `LaSalle_n.lean`—the $\dot{V}$ zero-equivalence condition, state and weight uniform boundedness, energy continuity, and compactness of the positively invariant set—were originally stated as axioms but are now proved as theorems using ISS-based Lyapunov arguments and ε - δ arguments.

Axioms. The full formalization relies on exactly 4 axioms, all of which are standard results in the dynamical systems literature:

1. `lasalle_convergence` — LaSalle’s principle, 2-node case (`LaSalle.lean:22`);
2. `lasalle_nonempty_intersection` — maximal invariant set nonempty (`LaSalle.lean:34`);
3. `lasalle_convergence_n` — LaSalle’s principle, n -node case (`LaSalle_n.lean:213`);
4. `energy_hasDerivAt_n` — chain rule for energy along ODE trajectories (`LaSalle_n.lean:506`).

All other claims—76 theorems and lemmas across 10 modules—are proved with complete Lean proofs, with zero `sorry` or `admit` statements.

Availability. The complete source code is available at:

<https://github.com/yohanchen1/lyapunov-cohesive-emergence>

A full build log (3,302 jobs, zero errors) is included in the supplementary material. A citable archived version will be deposited on Zenodo upon acceptance.

8. Conclusion

This paper has presented a rigorous Lyapunov-based analysis of cohesive structure emergence in sparse dynamic networks. The three theorems together articulate a clear mathematical narrative. Theorem 1 constructs a Lyapunov function whose monotonic decrease provides a global energy landscape guiding the system toward order. Theorem 2 proves that convergence is inescapable: every trajectory, regardless of initial condition, must approach the equilibrium set. Theorem 3 then reveals what structure these equilibria possess: a δ -cohesive partition of vertices into excitatory and inhibitory clusters, with weight factorization $w_{ij}^* \propto \phi_i \phi_j$ and a self-consistency equation $h(x_i^*) \propto \sum_{N(i)} \phi_j^2$ that encodes the collective activation of each node’s neighborhood. Together, these results verify the intuition that local energy-dissipative dynamics on a sparse network necessarily give rise to global cohesive structure.

Boundaries. This work is a proof of principle: it establishes that the core mechanism—coupled Hopfield-Hebbian dynamics on a fixed graph—is mathematically sufficient to produce δ -cohesive structure. However, the topology is fixed and the dynamics are deterministic. Future work should extend the analysis to growing or rewiring networks, stochastic perturbations, and systems with multiple interacting attractors. The connection between δ -cohesive partitions and k -plexes [16] suggests applications to formalizing neural assemblies, where a cohesive subgraph of strongly interconnected nodes could serve as the substrate for a distributed representation.

Future work. Extending the analysis to time-varying topologies, heterogeneous learning rates, and continuous attractor networks are natural next directions. The self-consistency equation (6) suggests a mechanism for competitive dynamics between cohesive clusters that may be relevant to understanding neural competition and decision-making. Eliminating the remaining 4 axioms listed in Section 7—in particular, proving `energy_hasDerivAt_n` as a Lean theorem using the existing chain-rule infrastructure—would bring the formalization to full self-contained status.

9. A Philosophical Conjecture

The results presented in this paper invite broader reflection. Beyond the mathematics, a philosophical hypothesis is ventured—not as a theorem, but as a direction for future investigation.

The conjecture. Knowledge exists in a fuzzy, sub-symbolic form that harbors latent structural conditions. Logic is not external to these structures but internalized within them. What is called “expression of knowledge” is a post-hoc translation of these internal cohesive structures into the sequential medium of language. In this view, understanding is not symbol manipulation but the self-organization of cohesive structures under internal constraints—and language is the aftermath of thought, not its medium.

What this is not. This is not a theorem of this paper. It is not proved, and it does not follow from the mathematics in any deductive sense. It is offered as an academic hypothesis that the mathematical framework makes plausible but does not establish. It is presented not as a conclusion but as an invitation: a possible reorientation of the relationship between structure, knowledge, and language.

A path toward testability. The mathematical framework developed in this paper may provide a foundation for rendering this conjecture empirically approachable. A natural next step would be to construct “concept networks”—graphs whose nodes represent primitive concepts and whose edges represent logical or associative relations—and evolve them under the dynamics (1)-(2). If the resulting δ -cohesive structures can be shown to correspond to meaningful logical groupings (e.g., nodes that participate in the same inference chain or belong to the same semantic category), then the conjecture moves from philosophical speculation toward testable scientific proposition. The self-consistency equation (6), in particular, offers a concrete mathematical foothold: it predicts that a node’s equilibrium activation is determined by the collective activation of its neighbors, which is precisely the kind of distributed, context-dependent binding that a “concept network” would require.

Relation to existing frameworks. This conjecture resonates with, but is distinct from, several established ideas. It shares with connectionism the view that cognition emerges from distributed interactions rather than centralized symbol processing. It differs from classical symbolic AI in treating language as an approximate readout of parallel internal states rather than the primary medium of thought. Dynamic systems approaches to cognition [17] also emphasize continuous-time dynamics; the present conjecture differs by proposing δ -cohesive partitions as a candidate organizing principle, though whether such structures arise in cognitive networks remains an open empirical question.

Invitation to critique. This conjecture is presented with full awareness of its speculative nature. The intention is not to make a sweeping claim but to identify a direction that the mathematical results of this paper make newly visible. Whether this direction proves fruitful is an empirical question that cannot be answered by mathematics alone. Critical examination, alternative formulations, and experimental tests are all welcome. The conjecture stands not as the conclusion of this work, but as a signpost pointing toward questions that are worth asking.

10. Acknowledgments

This work was inspired by a long-standing intuition about cohesive structures as fundamental units of organization in complex systems. The computational verification was enabled by Wolfram Engine, SciPy [12], and NetworkX [18]. The Lean 4 formalization was developed with assistance from the DeepSeek large language model, and builds upon the Mathlib mathematical library maintained by the Lean community.

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